REPRESENTABILITY OF THE DOUBLE DIAGONAL IS A CONDITION ON EQUALITIES OF MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Inertia as a double fibre product. categories-section-inertia (cross-chapter, named section). Identifies X x_{X x X} X with the inertia and the double diagonal with its identity section.

Source reference. Original reference 04Z2: focused-seeds/geometry/S9-crit-second-diagonal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{X}$ be a category fibred in groupoids over $(Sch/S)_{fppf}$. The following are equivalent*

(1) *$\Delta_\Delta : \mathcal{X} \to \mathcal{X} \times_{\mathcal{X}\times\mathcal{X}} \mathcal{X}$ is representable by algebraic spaces,*
(2) *for every 1-morphism $\mathcal{V} \to \mathcal{X} \times \mathcal{X}$ with $\mathcal{V}$ representable (by a scheme) the fibre product $\mathcal{Y} = \mathcal{X} \times_{\Delta,\mathcal{X}\times\mathcal{X}} \mathcal{V}$ has diagonal representable by algebraic spaces.*

Proof. Although this is a bit of a brain twister, it is completely formal. Namely, recall that $\mathcal{X} \times_{\mathcal{X}\times\mathcal{X}} \mathcal{X} = \mathcal{I}_\mathcal{X}$ is the inertia of $\mathcal{X}$ and that Δ_Δ is the identity section of $\mathcal{I}_\mathcal{X}$, see Categories, Section 04Z2. Thus condition (1) says the following: Given a scheme V, an object x of $\mathcal{X}$ over V, and a morphism $\alpha : x \to x$ of $\mathcal{X}_V$ the condition "$\alpha = \mathrm{id}_x$" defines an algebraic space over V. (In other words, there exists a monomorphism of algebraic spaces $W \to V$ such that a morphism of schemes $f : T \to V$ factors through W if and only if $f^*\alpha = \mathrm{id}_{f^*x}$.)

On the other hand, let V be a scheme and let x, y be objects of $\mathcal{X}$ over V. Then (x, y) define a morphism $\mathcal{V} = (Sch/V)_{fppf} \to \mathcal{X} \times \mathcal{X}$. Next, let $h : V' \to V$ be a morphism of schemes and let $\alpha : h^*x \to h^*y$ and $\beta : h^*x \to h^*y$ be morphisms of $\mathcal{X}_{V'}$. Then (α, β) define a morphism $\mathcal{V}' = (Sch/V)_{fppf} \to \mathcal{Y} \times \mathcal{Y}$. Condition (2) now says that (with any choices as above) the condition "$\alpha = \beta$" defines an algebraic space over V.

To see the equivalence, given (α, β) as in (2) we see that (1) implies that "$\alpha^{-1} \circ \beta = \mathrm{id}_{h^*x}$" defines an algebraic space. The implication (2) $\Rightarrow$ (1) follows by taking $h = \mathrm{id}_V$ and $\beta = \mathrm{id}_x$. □

Date: stacks-project:criteria.tex:local-snapshot-20261012; GNU FDL 1.2 or later.